Lecture 15

Metrics, the Hodge star and Maxwell's equations

Objectives. *A metric measures tangent vectors and selects a volume form once an orientation is chosen. The Hodge star then turns a form into a form of complementary degree. On spacetime this operation packages Maxwell's four equations into two equations for forms. The monopole flux shows why closed field forms need not have global potentials. On a closed oriented Riemannian manifold, the metric selects a harmonic representative of each de Rham class.*

By the end of this lecture you will be able to:

- Use a metric to lower indices and compute the Hodge star with orientation and signature specified.
- Derive all four Maxwell equations and charge conservation from the form equations.
- Use a nonzero sphere period to rule out a global monopole potential.
- Compute the codifferential and the harmonic representative of a form on the circle.

Unit 15A Metrics and the Hodge star

15.1 Metrics and the identification of vectors with covectors

Let M be a smooth n-manifold.

Definition 15.1. A *Riemannian metric* g is a smoothly varying positive-definite symmetric bilinear form g_p on each tangent space T_pM.

Definition 15.2. A *pseudo-Riemannian metric* requires nondegeneracy instead of positive definiteness. Nondegeneracy means that $g_p(v, w) = 0$ for all $w \in T_pM$ implies $v = 0$.

We use g for the metric throughout Part II.

By Sylvester's law of inertia, a nondegenerate symmetric form admits a basis with diagonal entries $+1$ and -1.

Definition 15.3. Such a basis is called *pseudo-orthonormal*. The *signature* is $(n - s, s)$, where s counts the negative entries. A *Lorentzian metric* in dimension $n \geq 2$ has signature $(1, n - 1)$.

The two counts are independent of the basis. This is the signature result imported from *Linear Algebra for Physics*, Lecture 11, the example *Signature of a form*. For a smooth nondegenerate metric, the signature is constant on each connected component: eigenvalues vary continuously and cannot cross zero.

In coordinates, with $i, j \in \{1, \dots, n\}$, $g = g_{ij}\mathrm{d}x^i \otimes \mathrm{d}x^j$. Its inverse matrix is written (g^{ij}).

Definition 15.4. For $X \in T_pM$ and $\alpha \in T_p^*M$, the mutually inverse *musical maps* are the maps

$$\flat \colon T_pM \to T_p^*M, \qquad \sharp \colon T_p^*M \to T_pM,$$

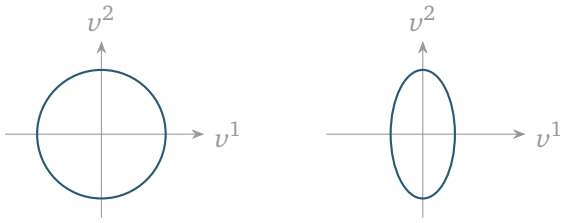
given by

$$X^\flat = g(X, \cdot), \qquad (X^\flat)_i = g_{ij}X^j, \qquad (\alpha^\sharp)^i = g^{ij}\alpha_j.$$

These maps lower and raise an index using a metric. The coordinate relabelling $x_i \mapsto x^i$ in Lecture 7 did not apply a metric. For example, in one tangent space take the following metric matrix and vector components. The resulting lower components are

$$(g_{ij}) = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \qquad (X^i) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad ((X^\flat)_i) = \begin{pmatrix} 4 \\ 7 \end{pmatrix}.$$

The vector has squared length $4+14 = 18$. The figure shows how a positive metric determines its unit vectors.



$g = \mathrm{diag}(1, 1)$ $\qquad$ $g = \mathrm{diag}(4, 1)$

Figure 15.1: At a point, Euclidean unit vectors form a circle. For $g = 4\mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y$ they form an ellipse with horizontal radius $1/2$.

Proposition 15.5 (Existence and induced metrics · *proved given partitions of unity*)**.** *Every smooth manifold admits a Riemannian metric. An immersion into a Riemannian manifold induces a Riemannian metric by pullback.*

Proof. On each coordinate domain use $g_a = \sum_i \mathrm{d}x_a^i \otimes \mathrm{d}x_a^i$. Choose a subordinate partition ρ_a from Theorem 12.8 and put $g = \sum_a \rho_a g_a$, extending each summand by zero. Supports make the extensions smooth. Local finiteness makes the sum smooth. For nonzero v, every summand $\rho_a g_a(v, v)$ is nonnegative, and at least one is positive because $\sum_a \rho_a = 1$. Thus the metric is positive definite. For an immersion f, the differential is injective, so $(f^*g)(v, v) = g(\mathrm{d}f(v), \mathrm{d}f(v)) > 0$ for $v \neq 0$. Smoothness and symmetry are preserved by pullback. ∎

Definition 15.6. An *isometry* is a diffeomorphism $f \colon (M, g_M) \to (N, g_N)$ with $f^*g_N = g_M$.

A pseudo-Riemannian metric can restrict degenerately: for the matrix $\mathrm{diag}(1, -1, -1, -1)$, the vector $(1, 1, 0, 0)$ spans a line on which the restricted form is zero. So the induced-metric assertion used positivity.

We compute the induced metric on the unit sphere using the parametrization

$$p(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \qquad 0 < \theta < \pi,$$

where ϕ lies in an angular chart. The coordinate tangents have dot products

$$\partial_\theta p \cdot \partial_\theta p = 1, \qquad \partial_\theta p \cdot \partial_\phi p = 0, \qquad \partial_\phi p \cdot \partial_\phi p = \sin^2\theta.$$

Thus the induced metric is

$$g = d\theta \otimes d\theta + \sin^2\theta \, d\phi \otimes d\phi.$$

The figure compares the lengths of the coordinate tangents at two points.

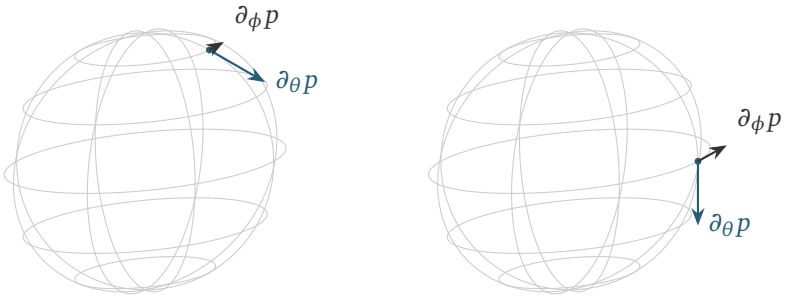


Figure 15.2: On the unit sphere at $\phi = 0$, the coordinate tangents are $\partial_\theta p = (\cos\theta, 0, -\sin\theta)$ and $\partial_\phi p = (0, \sin\theta, 0)$. Foreshortening in the receding y-direction preserves the ratio between the two $\partial_\phi p$ arrows, whose lengths on the sphere are $1/2$ at $\theta = \pi/6$ and 1 at $\theta = \pi/2$.

Definition 15.7. The *length* of a piecewise smooth curve $\gamma : [a, b] \to M$ in a Riemannian manifold is

$$\ell(\gamma) = \int_a^b \sqrt{g(\dot\gamma, \dot\gamma)}\, dt.$$

A meridian from the north pole to the south pole has length $\int_0^\pi 1 \, d\theta = \pi$. A latitude at $\theta = \theta_0 \in (0, \pi)$ has length $\int_0^{2\pi} \sin\theta_0 \, d\phi = 2\pi \sin\theta_0$. The meridian length includes the endpoints by continuity of the induced metric.

15.2 Volume forms and the Hodge star

Definition 15.8. On an oriented pseudo-Riemannian n-manifold, the *metric volume form* is, in positive coordinates,

$$\mu = \sqrt{|\det(g_{ij})|}\, dx^1 \wedge \cdots \wedge dx^n.$$

The defining normalization is $\mu = \theta^1 \wedge \cdots \wedge \theta^n$ when $\theta^1, \dots, \theta^n$ is dual to a positive pseudo-orthonormal basis.

To justify the coefficient, choose such a basis with metric matrix $S = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_n)$, where $\varepsilon_i = \pm 1$. If the coordinate basis has columns A in that basis, its metric matrix is $A^T S A$. Hence $|\det g| = (\det A)^2$. The dual bases satisfy $dx^1 \wedge \cdots \wedge dx^n = (\det A)^{-1} \theta^1 \wedge \cdots \wedge \theta^n$. Since $\det A > 0$, the coefficient gives the required normalization. Under a positive coordinate change the square root of the absolute metric determinant contributes the absolute Jacobian and the top wedge contributes its inverse. The expressions therefore agree. They are smooth because the determinant never vanishes. This is the chosen volume form μ of Lecture 11, now fixed by metric and orientation. Without orientation the same coefficient defines a density.

For the sphere metric just computed, $\sqrt{\det(g_{ij})} = \sin\theta$, so $\mu = \sin\theta\,\mathrm{d}\theta \wedge \mathrm{d}\phi$. This is the outward area form from Lecture 10.

In Lecture 10, a spatial vector field gave both a work 1-form and a flux 2-form. The metric identifies the vector with the work form. The metric and orientation then determine the flux form. The Hodge star will give this second passage in every dimension and degree.

For $0 \le k \le n$, the inverse metric gives a bilinear pairing of covectors. Its induced pairing on decomposable k-forms is

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_k, \beta_1 \wedge \cdots \wedge \beta_k \rangle_g = \det\left[g^{-1}(\alpha_i, \beta_j)\right]_{i,j=1}^{k},$$

extended bilinearly. The determinant is multilinear and alternating in each list of covectors, so it depends only on the two wedges. On functions it is ordinary multiplication. For example, in Euclidean three-space,

$$\langle \mathrm{d}x \wedge \mathrm{d}y, \mathrm{d}x \wedge \mathrm{d}y \rangle_g = \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1.$$

This uses Lecture 10's determinant wedge, with no additional normalization from the fermionic wedge of *Quantum Mechanics I*. The pairing is nondegenerate: in a pseudo-orthonormal dual basis the basis wedges are orthogonal and each has norm squared +1 or −1.

Definition 15.9. On an oriented pseudo-Riemannian n-manifold, the *Hodge star* is the pointwise real-linear map

$$\star\colon \Omega^k(M) \longrightarrow \Omega^{n-k}(M)$$

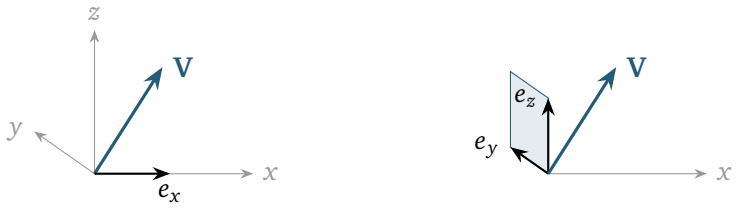
characterized by

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \mu \qquad (\alpha, \beta \in \Omega^k(M)).$$

In oriented Euclidean $\mathbb{R}^2$, the basis covectors are orthonormal. For example, $\mathrm{d}y \wedge (-\mathrm{d}x) = \mathrm{d}x \wedge \mathrm{d}y$ fixes the sign of $\star\mathrm{d}y$. The defining equation gives

$$\star 1 = \mathrm{d}x \wedge \mathrm{d}y, \qquad \star\mathrm{d}x = \mathrm{d}y, \qquad \star\mathrm{d}y = -\mathrm{d}x, \qquad \star(\mathrm{d}x \wedge \mathrm{d}y) = 1.$$

In Euclidean three-space, the star sends the work 1-form to the flux 2-form. The figure compares their evaluations for one vector.



$\mathbf{V}^\flat(e_x) = 2$ $\qquad$ $(\star\mathbf{V}^\flat)(e_y, e_z) = 2$

Figure 15.3: For $\mathbf{V} = (2, 1, 2)$, evaluation of $\mathbf{V}^\flat$ on e_x and evaluation of $\star\mathbf{V}^\flat$ on (e_y, e_z) both give 2. The segment measures a component along a path; the oriented square measures flux.

Theorem 15.10 (Hodge star). *Let (M, g) be an oriented pseudo-Riemannian n-manifold and $0 \le k \le n$. There is a unique pointwise real-linear operator, smooth in position, $\star\colon \Omega^k(M) \to \Omega^{n-k}(M)$ such that*

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \mu \qquad (\alpha, \beta \in \Omega^k(M)).$$

If the metric has s negative directions, then on k-forms

$$\star\star\alpha = (-1)^{k(n-k)+s}\alpha.$$

Proof. We prove existence and uniqueness by the wedge pairing at a point. We then compute the square on basis wedges. The wedge pairing between degrees k and $n-k$ is nondegenerate: pairing with a basis wedge selects the coefficient of its complementary wedge. It therefore represents the functional $\alpha \mapsto \langle \alpha, \beta \rangle_g \mu$ uniquely. The coordinate linear system uses smooth inverse-metric coefficients and nonzero volume coefficient, so its solution varies smoothly.

For the sign, use a positively oriented pseudo-orthonormal dual basis $\theta^1, \dots, \theta^n$ at one point. For an increasing index set I, write θ^I for its wedge, I^c for its ordered complement, $\varepsilon_I = \prod_{i \in I} \varepsilon_i$, and σ_I for the sign of the permutation (I, I^c). The defining equation gives

$$\star\theta^I = \varepsilon_I \sigma_I \theta^{I^c}.$$

The two metric signs multiply to $\prod_i \varepsilon_i = (-1)^s$. Interchanging the blocks I and I^c uses $k(n-k)$ transpositions, so $\sigma_I \sigma_{I^c} = (-1)^{k(n-k)}$. Applying the formula twice proves the square identity on every basis wedge, hence on every form. ∎

For example, in oriented Euclidean $\mathbb{R}^3$, $\star dx = dy \wedge dz$ and $\star(dy \wedge dz) = dx$. The complementary plane is shown in the figure.

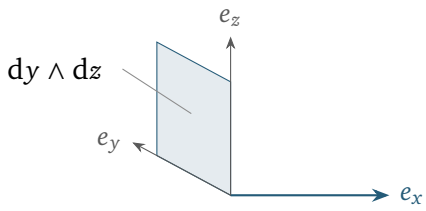


Figure 15.4: In Euclidean three-space, the metric identifies the covector dx with e_x. Its star is the oriented area form $dy \wedge dz$ on the complementary plane.

Reversing orientation negates μ and therefore negates the star. For a compactly supported real form α, the integral $\int_M \alpha \wedge \star\alpha$ is unchanged: the integrand and the integration orientation both change sign. The pairing need not be positive in indefinite signature. For complex forms extend the star complex-linearly and use $\bar{\alpha} \wedge \star\beta$ for the Hermitian pairing, antilinear in its first argument.

15.2.1 The spacetime star

Use coordinates $(x^0, x^1, x^2, x^3) = (t, x, y, z)$ on an open subset of spacetime $\mathbb{R}^4$. Greek indices run from 0 to 3 and spatial indices from 1 to 3. Take the Minkowski metric and orientation

$$\eta = \operatorname{diag}(1, -1, -1, -1), \qquad \mu = dt \wedge dx \wedge dy \wedge dz.$$

This is the metric recalled from *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Units have $c = 1$ and rationalized sources. The metric has $s = 3$, so $\star^2 = -1$ on 2-forms. A real 2-form ω satisfying $\star\omega = \omega$ or $\star\omega = -\omega$ must therefore be zero. This distinguishes the Lorentzian fields here from the Euclidean self-dual fields of Lecture 23. Euclidean four-space instead has $\star^2 = +1$ on 2-forms.

The six basis 2-forms give a direct way to use the signature convention. For example, $dt \wedge dx$ has norm squared -1 and $(dt \wedge dx) \wedge (dy \wedge dz) = \mu$. Its star is therefore $-dy \wedge dz$. The pairings used here are

$$\langle dt \wedge dx, dt \wedge dx \rangle_\eta = \det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1, \qquad \langle dy \wedge dz, dy \wedge dz \rangle_\eta = 1.$$

The same rule gives the table:

α	$\star\alpha$	α	$\star\alpha$
$dt \wedge dx$	$-dy \wedge dz$	$dy \wedge dz$	$dt \wedge dx$
$dt \wedge dy$	$-dz \wedge dx$	$dz \wedge dx$	$dt \wedge dy$
$dt \wedge dz$	$-dx \wedge dy$	$dx \wedge dy$	$dt \wedge dz$

The degree-one stars are:

$$\star dt = dx \wedge dy \wedge dz, \quad \star dx = dt \wedge dy \wedge dz,$$
$$\star dy = dt \wedge dz \wedge dx, \quad \star dz = dt \wedge dx \wedge dy.$$

Self-check. In oriented Euclidean $\mathbb{R}^3 \setminus \{0\}$, put $r = \sqrt{x^2 + y^2 + z^2}$ and take $\mathbf{E}^\flat = (x\,dx + y\,dy + z\,dz)/r^3$. Compute $\star\mathbf{E}^\flat$. Use its outward sphere period to decide whether it has a global 1-form primitive.

Answer. The Euclidean rules give

$$\star\mathbf{E}^\flat = \frac{x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy}{r^3}.$$

This is the Coulomb flux form of Lecture 12, written here as β_C. Its outward unit-sphere period is 4π, so Stokes rules out a global primitive. Here the spatial Euclidean star is used, not the four-dimensional spacetime star.

Unit 15B Electromagnetic fields and global potentials

15.3 Maxwell's equations as equations for forms

Physical input. We use classical electromagnetism in vacuum, with smooth electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, charge density ρ and spatial current $\mathbf{J}$. Electric work per unit charge along a spatial path is $\int E_i dx^i$. Magnetic flux through an oriented spatial surface is $\int \mathbf{B} \cdot \mathbf{n}\, dS$. Charge in a spatial region is the integral of ρ over its volume. The outward current flux measures charge leaving per unit time. In the units specified above, we take the vacuum Maxwell laws with sources as physical input:

$$\operatorname{div}\mathbf{B} = 0, \quad \partial_t\mathbf{B} + \operatorname{curl}\mathbf{E} = 0, \qquad \operatorname{div}\mathbf{E} = \rho, \quad \operatorname{curl}\mathbf{B} - \partial_t\mathbf{E} = \mathbf{J}.$$

This is the electromagnetic model of Nakahara, §7.9, Example 7.16, expressed with our signature and source convention.

The Cartesian spatial components use the Euclidean spatial metric. They are not obtained by lowering spacetime indices with η. For a spatial vector V, we write its flux 2-form as

$$\mathcal{B}(V) = V_x \mathrm{d}y \wedge \mathrm{d}z + V_y \mathrm{d}z \wedge \mathrm{d}x + V_z \mathrm{d}x \wedge \mathrm{d}y.$$

Magnetic flux integrates a 2-form over a spatial surface. Electric work integrates a 1-form over a spatial path. Faraday's law relates the change of magnetic flux to electric circulation. On a spacetime wall swept out by a path, the electric term therefore has both a spatial differential and a time differential.

Definition 15.11. The real *field-strength 2-form* is

$$F_{\mathrm{em}} = E_i \mathrm{d}x^i \wedge \mathrm{d}t + B_x \mathrm{d}y \wedge \mathrm{d}z + B_y \mathrm{d}z \wedge \mathrm{d}x + B_z \mathrm{d}x \wedge \mathrm{d}y.$$

Definition 15.12. The *current 1-form* and *current 3-form* are

$$j = \rho \mathrm{d}t - J_i \mathrm{d}x^i, \qquad \mathcal{J} = \star j.$$

The subscripts em distinguish these real electromagnetic forms from later connection forms. The current 1-form is the metric dual of the charge-current vector:

$$j = (\rho \partial_t + J_i \partial_i)^\flat.$$

On a fixed-time surface, F_{em} restricts to $\mathcal{B}(\mathbf{B})$. It also satisfies $F_{\mathrm{em}}(\partial_i, \partial_t) = E_i$. For a compact fixed oriented surface Σ, put $\Phi(t) = \int_\Sigma \mathcal{B}(\mathbf{B}(t))$. For $t_1 < t_2$, orient $[t_1, t_2] \times \Sigma$ by $\mathrm{d}t$ followed by the orientation of Σ. We assume that this cylinder lies in the field domain. Let s be positive arclength along $\partial\Sigma$. The outward-first boundary rule gives

$$\partial([t_1, t_2] \times \Sigma) = \Sigma_{t_2} - \Sigma_{t_1} - [t_1, t_2] \times \partial\Sigma.$$

On the wall with product orientation, the electric term is $-\mathrm{d}t \wedge E_i \mathrm{d}x^i$. The induced wall orientation contributes a second minus sign. The figure records the resulting signed integrals.

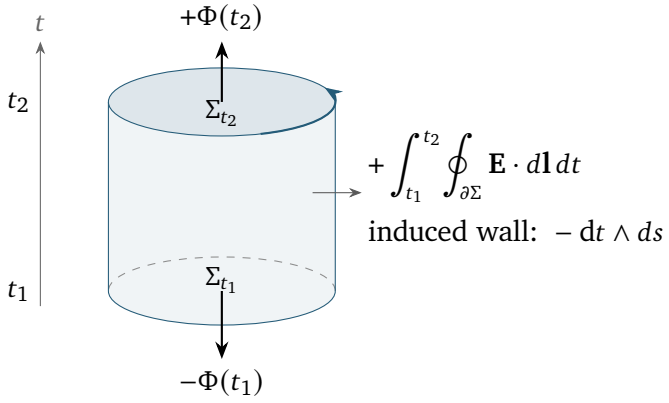


Figure 15.5: The spacetime cylinder over a positively oriented disc has top flux $+\Phi(t_2)$ and bottom flux $-\Phi(t_1)$. The induced wall contribution is $+\int_{t_1}^{t_2} \oint_{\partial\Sigma} \mathbf{E} \cdot d\mathbf{l}\, dt$, with $\partial\Sigma$ counter-clockwise viewed from above.

Stokes therefore turns the homogeneous form equation into integrated Faraday's law:

$$0 = \int_{\partial([t_1,t_2]\times\Sigma)} F_{\mathrm{em}} = \Phi(t_2) - \Phi(t_1) + \int_{t_1}^{t_2} \oint_{\partial\Sigma} \mathbf{E}\cdot d\mathbf{l}\, dt.$$

Theorem 15.13 (Maxwell's equations in forms)**.** *On the specified spacetime, the Maxwell laws stated as physical input are equivalent to*

$$\mathrm{d}F_{\mathrm{em}} = 0, \qquad \mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}.$$

These equations imply charge conservation, $\partial_t\rho + \operatorname{div}\mathbf{J} = 0$. When these equations hold, every point has a neighbourhood with $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ for a real 1-form A_{em}.

Proof. The spacetime star table gives

$$\begin{aligned}
{\star}F_{\mathrm{em}} &= B_i\mathrm{d}t\wedge\mathrm{d}x^i + E_x\mathrm{d}y\wedge\mathrm{d}z + E_y\mathrm{d}z\wedge\mathrm{d}x + E_z\mathrm{d}x\wedge\mathrm{d}y,\\
\mathcal{J} &= \rho\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z - J_x\mathrm{d}t\wedge\mathrm{d}y\wedge\mathrm{d}z + J_y\mathrm{d}t\wedge\mathrm{d}x\wedge\mathrm{d}z - J_z\mathrm{d}t\wedge\mathrm{d}x\wedge\mathrm{d}y.
\end{aligned}$$

In the flux notation, the source form is

$$\mathcal{J} = \rho\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z - \mathrm{d}t\wedge\mathcal{B}(\mathbf{J}).$$

Differentiation gives

$$\begin{aligned}
\mathrm{d}F_{\mathrm{em}} &= (\operatorname{div}\mathbf{B})\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z + \mathrm{d}t\wedge\mathcal{B}(\partial_t\mathbf{B} + \operatorname{curl}\mathbf{E}),\\
\mathrm{d}{\star}F_{\mathrm{em}} &= (\operatorname{div}\mathbf{E})\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z + \mathrm{d}t\wedge\mathcal{B}(\partial_t\mathbf{E} - \operatorname{curl}\mathbf{B}).
\end{aligned}$$

For instance, the coefficient of $\mathrm{d}t\wedge\mathrm{d}y\wedge\mathrm{d}z$ in the first equation is $\partial_t B_x + \partial_y E_z - \partial_z E_y$. The other cyclic coefficients give the other components of the curl. The magnetic term in the star contributes

$$\mathrm{d}(B_i\mathrm{d}t\wedge\mathrm{d}x^i) = \partial_j B_i\mathrm{d}x^j\wedge\mathrm{d}t\wedge\mathrm{d}x^i = -\mathrm{d}t\wedge\mathcal{B}(\operatorname{curl}\mathbf{B}).$$

The minus sign comes from moving $\mathrm{d}x^j$ past $\mathrm{d}t$. Independence of the four basis 3-forms gives the homogeneous equations. For the source equation the same comparison gives

$$\operatorname{div}\mathbf{E} = \rho, \qquad \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B} = -\mathbf{J}.$$

The second equality is the stated Ampère–Maxwell law. Applying d to the source equation gives $\mathrm{d}\mathcal{J} = 0$. Differentiating the displayed current form gives $\mathrm{d}\mathcal{J} = (\partial_t\rho + \operatorname{div}\mathbf{J})\mu$, which proves conservation. Finally choose a star-shaped coordinate neighbourhood and apply Theorem 13.11 to the closed 2-form. Its primitive is the required local potential. ∎

15.3.1 Gauss's law, charge conservation and covariance

On a fixed-time slice, $\mathrm{d}t$ pulls back to zero. The forms ${\star}F_{\mathrm{em}}$ and $\mathcal{J}$ restrict to $\mathcal{B}(\mathbf{E})$ and $\rho\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$. For a compact smooth spatial region V in this slice, Stokes gives Gauss's law:

$$\int_{\partial V} \mathbf{E}\cdot\mathbf{n}\, dS = \int_V \rho\, dx\, dy\, dz.$$

Thus electric flux through the outward boundary equals the enclosed charge. The integral of $\mathcal{J}$ over an oriented spacetime hypersurface measures signed net charge crossing that

hypersurface. Write $Q(t) = \int_V \rho(t)\,dx\,dy\,dz$ for a fixed region V. Assume that $[t_1, t_2] \times V$ lies in the field domain. Applying Stokes to $\mathrm{d}\mathcal{J} = 0$ on this cylinder gives

$$Q(t_2) - Q(t_1) + \int_{t_1}^{t_2} \int_{\partial V} \mathbf{J} \cdot \mathbf{n}\, dS\, dt = 0.$$

Positive outward current therefore decreases the charge in V. The charge cylinder shows the same boundary signs for a current 3-form.

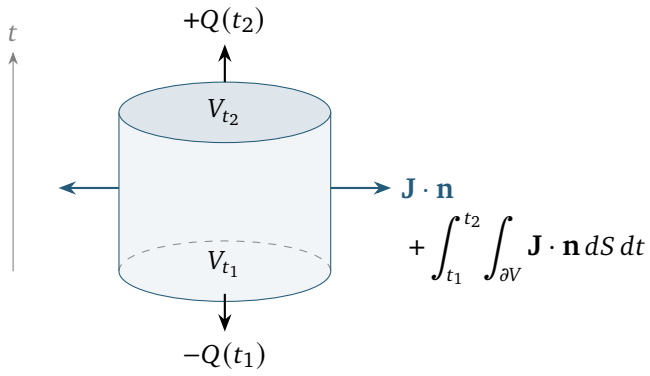


Figure 15.6: The top and bottom of a swept spatial region contribute $+Q(t_2)$ and $-Q(t_1)$, while the wall contributes positive outward current. This is a schematic with the third spatial direction suppressed.

The homogeneous equation $\mathrm{d}F_{\mathrm{em}} = 0$ needs neither a metric nor an orientation. The source equation uses the metric and orientation through the star. An orientation-preserving isometry f preserves the form pairing and volume. Pulling back the defining star identity and using uniqueness gives $f^*(\star\alpha) = \star(f^*\alpha)$. Pullback also commutes with d by Lecture 10. Hence f^*F_{em} and $f^*\mathcal{J}$ satisfy the same form equations. In particular, proper Lorentz transformations preserve Maxwell's equations. Their pullbacks mix the time-space and space-space components of F_{em}, which are its electric and magnetic parts.

15.3.2 Potentials and gauge transformations

Write the local potential in terms of its scalar and spatial parts as

$$A_{\mathrm{em}} = -\phi \mathrm{d}t + A_i \mathrm{d}x^i.$$

Taking its derivative gives $E_i = -\partial_i\phi - \partial_t A_i$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$, where $\mathbf{A} = (A_x, A_y, A_z)$.

Definition 15.14. For a smooth real function χ, a *gauge transformation* is $A_{\mathrm{em}} \mapsto A_{\mathrm{em}} + \mathrm{d}\chi$.

This transformation leaves F_{em} unchanged because $\mathrm{d}^2\chi = 0$. Its component form is $A_i \mapsto A_i + \partial_i\chi$ and $\phi \mapsto \phi - \partial_t\chi$. This recalls *Theoretical Mechanics*, Lecture 3, *The charged particle*: the same transformation changes the particle Lagrangian by a total derivative.

The Landau–Lifshitz lower potential has the opposite sign at matched units.[1]

[1] Thus $A^{\mathrm{LL}} = -A_{\mathrm{em}}$ and $F^{\mathrm{LL}} = -F_{\mathrm{em}}$. The subsection *Conventions in electromagnetic sources* at the end of §15.4 translates the source equations of Nakahara and Landau–Lifshitz.

15.4 The monopole obstruction

Let $r = \sqrt{x^2 + y^2 + z^2}$ and let $\Phi_{\mathrm{m}} \in \mathbb{R} \setminus \{0\}$ be the magnetic flux. We consider a static *magnetic monopole* field on $\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$. Its magnetic field is $\mathbf{B} = \Phi_{\mathrm{m}}(x, y, z)/(4\pi r^3)$. Its electric field is zero. Its field 2-form is

$$F_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi} \frac{x\,\mathrm{d}y \wedge \mathrm{d}z + y\,\mathrm{d}z \wedge \mathrm{d}x + z\,\mathrm{d}x \wedge \mathrm{d}y}{r^3}.$$

Proposition 15.15 (No global monopole potential · *proved given partitions of unity*)**.** *This form is closed and has period Φ_{m} on the outward unit sphere in each fixed-time slice. It has local potentials but no global 1-form potential on $\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$, spacetime with the monopole worldline removed.*

Proof. The form is time-independent, with no time differential. Its spatial part is the Coulomb flux form of §12.7 times $\Phi_{\mathrm{m}}/(4\pi)$. Lecture 12 proved closedness and sphere period 4π for that form, giving the stated period. Local potentials follow from the Poincaré lemma. If $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ globally, restriction to a fixed-time sphere and Stokes would give zero period, contradicting $\Phi_{\mathrm{m}} \neq 0$. ∎

This is a conditional field model; no existence of a physical monopole is assumed. The nonzero sphere period says that $[F_{\mathrm{em}}] \neq 0$ in $H^2_{\mathrm{dR}}(\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\}))$. It is the Coulomb cohomology class computed in Lecture 13, with the stated scaling. The figure shows the removed worldline and a sphere that detects the class.

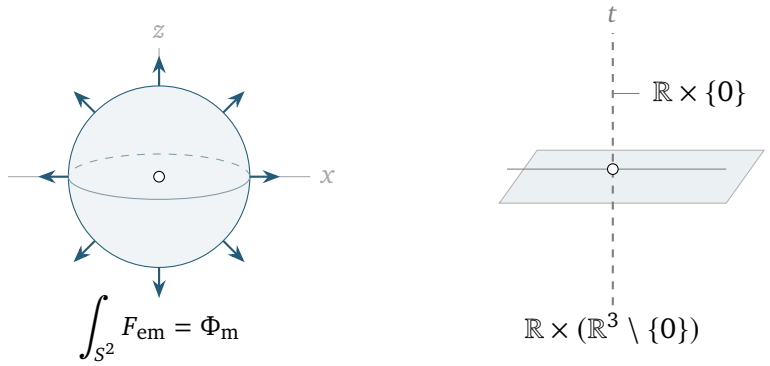


Figure 15.7: The omitted set is the whole time axis through the spatial origin. Each outward fixed-time sphere has flux Φ_{m}, with the arrows shown for positive flux and reversed when $\Phi_{\mathrm{m}} < 0$.

The obstruction prevents describing the field by one smooth potential on its whole domain. Local potentials must instead be related on their overlaps. No integer condition has been imposed on Φ_{m}. Lecture 22 studies the additional condition from a bundle and a charged particle.

15.4.1 Vacuum duality

Example 15.16 (An electric field and its magnetic dual)**.** Take a constant electric field $\mathbf{E} = E_0 e_x$, with $E_0 \neq 0$, and zero magnetic field on Minkowski space. Its potential and field are

$$A_{\mathrm{em}} = E_0 x\,\mathrm{d}t, \qquad F_{\mathrm{em}} = E_0\,\mathrm{d}x \wedge \mathrm{d}t.$$

All field derivatives vanish, so Maxwell's equations hold with zero sources. The table gives $\star F_{\mathrm{em}} = E_0 \mathrm{d}y \wedge \mathrm{d}z$. This is the field of a constant magnetic field $\mathbf{B}' = E_0 e_x$ and zero electric field. A potential for it is $E_0 y \mathrm{d}z$. Thus the transformation $F_{\mathrm{em}} \mapsto \star F_{\mathrm{em}}$ changes this electric solution into a magnetic solution.

Definition 15.17. For a source-free field, the transformation $F_{\mathrm{em}} \mapsto \star F_{\mathrm{em}}$ is *vacuum duality*.

Indeed, $\mathrm{d}F_{\mathrm{em}} = 0$ and $\mathrm{d}\star F_{\mathrm{em}} = 0$ imply $\mathrm{d}(\star F_{\mathrm{em}}) = 0$ and $\mathrm{d}\star(\star F_{\mathrm{em}}) = -\mathrm{d}F_{\mathrm{em}} = 0$. The transformed components are $\mathbf{E}' = -\mathbf{B}$ and $\mathbf{B}' = \mathbf{E}$. Applying duality twice gives $-F_{\mathrm{em}}$, rather than F_{em}.

Example 15.18 (The Coulomb field and the monopole). On $\mathbb{R} \times (\mathbb{R}^3 \setminus \{0\})$, let $Q \in \mathbb{R} \setminus \{0\}$ and take

$$\mathbf{E} = \frac{Q(x,y,z)}{4\pi r^3}, \qquad \mathbf{B} = 0, \qquad \phi = \frac{Q}{4\pi r}.$$

This static electric field has global potential $A_{\mathrm{em}} = -\phi \mathrm{d}t$. Indeed, $-\partial_i \phi = Q x^i/(4\pi r^3)$, so $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}} = E_i \mathrm{d}x^i \wedge \mathrm{d}t$. The field is source-free on this domain, since the spatial divergence and curl of $\mathbf{E}$ vanish away from the origin. Its spacetime star is

$$\star F_{\mathrm{em}} = \frac{Q}{4\pi} \frac{x\mathrm{d}y \wedge \mathrm{d}z + y\mathrm{d}z \wedge \mathrm{d}x + z\mathrm{d}x \wedge \mathrm{d}y}{r^3}.$$

This is the monopole form with $\Phi_{\mathrm{m}} = Q$. On a fixed-time outward sphere, the period of F_{em} is zero because $\mathrm{d}t$ pulls back to zero. The period of $\star F_{\mathrm{em}}$ is Q. Thus vacuum duality can take an exact field to a closed field with no global potential.

Conventions in electromagnetic sources

Source signs must be translated together. Nakahara's Example 7.16 uses signature $-+++$. For fixed orientation, metric reversal multiplies the k-form pairing and its star by $(-1)^k$. Thus the star on 2-forms is unchanged, while Nakahara's current is $j_{\mathrm{src}} = -j$. It follows that $\star_{\mathrm{src}} j_{\mathrm{src}} = \mathcal{J}$. His source equation is $\mathrm{d}^\dagger F_{\mathrm{em}} = j_{\mathrm{src}}$. Definition 7.6 gives $\mathrm{d}^\dagger = \star_{\mathrm{src}} \mathrm{d} \star_{\mathrm{src}}$ on Lorentzian 2-forms. Apply the source star and use its square $+1$ on 3-forms to obtain $\mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}$.

The lower potential $\phi \mathrm{d}t - A_i \mathrm{d}x^i$ of Landau–Lifshitz has the opposite sign to A_{em} at matched units. In these units, $A^{\mathrm{LL}} = -A_{\mathrm{em}}$ and $F^{\mathrm{LL}} = -F_{\mathrm{em}}$. To convert instead to Gaussian fields with $c = 1$, set $F_{\mathrm{G}} = \sqrt{4\pi} F_{\mathrm{em}}$ and $\mathcal{J}_{\mathrm{G}} = \mathcal{J}/\sqrt{4\pi}$. Then $\mathrm{d}\star F_{\mathrm{G}} = 4\pi \mathcal{J}_{\mathrm{G}}$. The Gaussian lower-potential convention $F_{\mathrm{G}}^{\mathrm{LL}} = -F_{\mathrm{G}}$ therefore gives $\mathrm{d}\star F_{\mathrm{G}}^{\mathrm{LL}} = -4\pi \mathcal{J}_{\mathrm{G}}$. This converts the field sign and source normalization together.

Self-check. On Minkowski space, take

$$F_{\mathrm{em}} = (2t + 3x)\mathrm{d}y \wedge \mathrm{d}z + (5y - t)\mathrm{d}z \wedge \mathrm{d}x + 7z\mathrm{d}x \wedge \mathrm{d}y - 2y\mathrm{d}x \wedge \mathrm{d}t.$$

Compute $\star F_{\mathrm{em}}$ and the coefficient of $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ in $\mathrm{d}F_{\mathrm{em}}$. Decide whether this field can satisfy Maxwell's equations for any electric sources.

Answer. The table gives

$$\star F_{\mathrm{em}} = (2t + 3x)\mathrm{d}t \wedge \mathrm{d}x + (5y - t)\mathrm{d}t \wedge \mathrm{d}y + 7z\mathrm{d}t \wedge \mathrm{d}z - 2y\mathrm{d}y \wedge \mathrm{d}z.$$

The required coefficient is $\partial_t B_x + \partial_y E_z - \partial_z E_y = 2$. In fact, $\operatorname{div}\mathbf{B} = 3 + 5 + 7 = 15$ and $\partial_t \mathbf{B} + \operatorname{curl}\mathbf{E} = (2, -1, 2)$. Thus $\mathrm{d}F_{\mathrm{em}} \neq 0$, independently of the electric source form.

Unit 15C Harmonic representatives

15.5 Harmonic representatives

A de Rham class consists of closed forms differing by exact forms. A manifold is *closed* if it is compact and has no boundary. On a closed oriented Riemannian manifold, we can compare the lengths of these forms using an integrated inner product. Harmonic representatives minimize this norm.

Definition 15.19. For real forms of equal degree on a closed oriented Riemannian manifold, the *integrated pairing* and its norm are

$$(\alpha,\beta)_{L^2}=\int_M \alpha\wedge\star\beta, \qquad \|\alpha\|_{L^2}^2=(\alpha,\alpha)_{L^2}.$$

Definition 15.20. The *codifferential* $\delta\colon \Omega^k(M)\to\Omega^{k-1}(M)$ is the formal adjoint of exterior differentiation, characterized by $(\mathrm{d}\alpha,\beta)_{L^2}=(\alpha,\delta\beta)_{L^2}$. We set it to zero on functions.

This operator differs from the connecting map δ_{MV} of Lecture 13. We first compute this adjoint on the unit circle $\mathbb{R}/(2\pi\mathbb{Z})$, with metric $\mathrm{d}\theta\otimes\mathrm{d}\theta$ and positive form $\mathrm{d}\theta$. The angle θ is local, but $\mathrm{d}\theta$ is a global 1-form. The star gives $\star 1=\mathrm{d}\theta$ and $\star\mathrm{d}\theta=1$. For smooth periodic functions f and a, integration by parts gives

$$(\mathrm{d}f,a\mathrm{d}\theta)_{L^2}=\int_0^{2\pi} f'a\,d\theta=-\int_0^{2\pi} fa'\,d\theta.$$

Hence $\delta(a\mathrm{d}\theta)=-a'$.

On a closed oriented Riemannian n-manifold, Stokes gives the formula on k-forms β, for $1\le k\le n$:

$$\delta\beta=(-1)^{n(k+1)+1}\star\mathrm{d}\star\beta.$$

To derive it, take $\alpha\in\Omega^{k-1}(M)$. The graded product rule and the absence of boundary give

$$0=\int_M \mathrm{d}(\alpha\wedge\star\beta)=(\mathrm{d}\alpha,\beta)_{L^2}+(-1)^{k-1}\int_M \alpha\wedge\mathrm{d}\star\beta.$$

Thus $\star\delta\beta=(-1)^k\mathrm{d}\star\beta$. Applying the star and using its square on degree $k-1$ gives the printed formula. In signature with s negative directions, the corresponding formal-adjoint formula has sign $(-1)^{n(k+1)+s+1}$. We use positivity only in Riemannian signature.

Definition 15.21. The *Hodge Laplacian* on $\Omega^k(M)$ is

$$\Delta=\mathrm{d}\delta+\delta\mathrm{d}.$$

Definition 15.22. A form is *harmonic* if $\Delta\alpha=0$.

Definition 15.23. A form is *coclosed* if $\delta\alpha=0$.

The operators d, δ and Δ act on forms. They are distinct from coordinate partial derivatives. On the circle the formulas above give

$$\Delta f=-f'', \qquad \Delta(a\mathrm{d}\theta)=-a''\mathrm{d}\theta.$$

A periodic function with zero second derivative is constant. Thus the harmonic functions are constants, and the harmonic 1-forms are constant multiples of $d\theta$. A 1-form with period P has harmonic representative $(P/(2\pi))d\theta$. Indeed, subtracting this constant form leaves a periodic coefficient of mean zero, whose integral from 0 to θ is a periodic primitive.

The Riemannian pointwise pairing is positive definite. The integrated norm therefore vanishes only for the zero form. Adjointness gives

$$(\Delta\alpha, \alpha)_{L^2} = \|d\alpha\|_{L^2}^2 + \|\delta\alpha\|_{L^2}^2.$$

Hence a form is harmonic exactly when it is both closed and coclosed. In particular, a harmonic form represents a de Rham class. For a harmonic k-form α with $k \geq 1$ and any $\gamma \in \Omega^{k-1}(M)$, adjointness gives $(\alpha, d\gamma)_{L^2} = (\delta\alpha, \gamma)_{L^2} = 0$. Consequently,

$$\|\alpha + d\gamma\|_{L^2}^2 = \|\alpha\|_{L^2}^2 + \|d\gamma\|_{L^2}^2.$$

Equality with $\|\alpha\|_{L^2}^2$ holds only when $d\gamma = 0$. Thus α is the unique form of minimum norm in its class. For degree zero there are no nonzero exact forms, so uniqueness in a class is immediate. This argument proves uniqueness when a harmonic representative exists.

Theorem 15.24 (Hodge representatives · *cited*)**.** *Every de Rham class on a closed oriented Riemannian manifold has a unique harmonic representative.*

The analytic existence theorem is cited from Nakahara, Theorem 7.8, §7.9.6. The closedness hypothesis is inherited there from Theorem 7.7. The statement does not apply to the Lorentzian spacetime used above.

On the flat torus $T^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$ with metric $dx \otimes dx + dy \otimes dy$ and orientation $dx \wedge dy$, the forms dx and dy descend from the plane. The coordinates x and y are not global real functions on the torus. The Euclidean star rules show that both forms are closed and coclosed. Their periods on the positive coordinate circles are $(2\pi, 0)$ and $(0, 2\pi)$. These periods distinguish their cohomology classes. On the round sphere, the metric volume form is also harmonic: its exterior derivative is zero by degree, and its star is the constant function 1.

Self-check. State in one sentence what a harmonic representative is and why we introduced it. On the unit circle, identify the harmonic representative of $(3 + 2\cos(2\theta) - 5\sin(3\theta))d\theta$ and justify that it is in the same class.

Answer. A harmonic representative is a closed and coclosed form selected by the metric as the unique form of minimum integrated norm in a de Rham class. The given form has period 6π, so its harmonic representative is $3d\theta$. The decisive exactness calculation is

$$(2\cos(2\theta) - 5\sin(3\theta))d\theta = d\Big(\sin(2\theta) + \frac{5}{3}\cos(3\theta)\Big).$$

The primitive is periodic, and $\delta(3d\theta) = 0$.

Summary. A Riemannian metric exists by partitions of unity and identifies tangent vectors with covectors. A metric and orientation define volume and the Hodge star. Its square depends on degree, dimension and the number of negative metric directions. With signature $+---$, Maxwell's equations are $\mathrm{d}F_{\mathrm{em}} = 0$ and $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. They imply charge conservation and local potentials. Monopole flux prevents a global potential. On closed oriented Riemannian manifolds, the cited Hodge theorem selects one harmonic form in each cohomology class. Lecture 16 uses a metric to differentiate tangent fields and define geodesics.

Reading. Nakahara, *Geometry, Topology and Physics*, second edition, §7.1 and §7.9, especially (7.172)–(7.177), Example 7.16, and Theorems 7.7–7.8. Tu, *Differential Geometry*, Theorem 1.12, pp. 6–7, and §16.4–§16.5, pp. 132–135.